

User Manual: Environmental Microorganism Two-Stage Intelligent Analysis System

1. Overview

This software is a two-stage cascaded deep learning application designed for the automated segmentation and taxonomic classification of environmental microorganisms (EMs) from microscopic slide images, benchmarked on the EMDS-6 dataset.

- **Stage 1 (Segmentation):** A U-Net architecture extracts the organism's morphological boundaries and filters out background noise, debris, and fluid impurities.
- **Stage 2 (Classification):** A ConvNeXt-Tiny classifier processes the masked foreground region to predict the taxonomic class along with Top-3 probability confidence scores.

2. Directory Structure & Execution Prerequisite

CRITICAL REQUIREMENT: The execution script and the pre-trained model weights must reside in the same working directory.

Ensure your directory contains the following files before launching:

— run_app.py	# Main desktop application script
— best_seg_unet.pth	# Trained Stage 1 U-Net model weights
— best_cls_model.pth	# Trained Stage 2 ConvNeXt model weights

Hardware Compatibility & Execution

- **CUDA GPU Accelerated:** Automatically engaged if an NVIDIA GPU with CUDA support is detected, delivering low-latency real-time inference.
- **CPU Compatible:** If no compatible GPU is found, the system automatically redirects tensor operations to system memory and CPU cores via internal map_location remapping. No manual configuration is required.

Launch command from your terminal or command prompt:

```
python run_app.py
```

3. How to Use

1. Launch the application. Verify that the bottom status bar displays 'System Ready' along with your active hardware device (CUDA or CPU).
2. Click the blue 'Open Microscopic Image' button located at the top-right corner.
3. Select a supported microscopic image (.png, .jpg, .jpeg, .bmp, .tif).
4. The system executes end-to-end inference and updates the three display panels:

- **Panel 1 (Raw Input):** Shows the original uploaded slide thumbnail.
- **Panel 2 (Segmented Foreground):** Displays the isolated organism with non-target background pixels set to zero (black).
- **Panel 3 (Predictions & Confidence):** Displays the Top-1 to Top-3 predicted taxonomic names and visual confidence progress bars.

4. System Constraints & Operational Boundaries

4.1. Closed-Set Classification Constraint (21 Predefined Classes)

- **Predefined Target Scope:** The model is trained strictly on the 21 microorganism categories provided by EMDS-6: Actinophrys, Arcella, Aspidisca, Codoniga, Colpoda, Epistylis, Euglypha, Paramecium, Rotifera, Vorticella, Noctiluca, Ceratium, Stentor, Siproostomum, Keratella Quadrala, Euglena, Gymnodinium, Gonyaulax, Phacus, Stylongchia, Synchaeta.
- **Out-of-Distribution (OOD) Behavior:** The model uses a standard normalized Softmax output layer. If an unmodeled organism, sludge particle, or fiber fragment is provided, the system will still force a prediction into one of the 21 classes. A high confidence percentage for an unknown specimen indicates visual resemblance to a training category rather than true taxonomic identity.

4.2. Image Resolution, Sharpness & Depth-of-Field

- **Resolution:** Input images should maintain a spatial resolution of at least 224×224 pixels. Low-resolution inputs risk detail loss during downsampling.
- **Focal Plane & Blur:** Microscopic specimens possess volumetric depth. Images with severe out-of-focus blur, motion artifacts, or poor contrast will disrupt Stage 1 U-Net boundary delineation, directly degrading the downstream masked classification input.

4.3. Spatial Composition & Multi-Target Density

- **Single-Organism Centering:** The pipeline is optimized for individual specimens situated near the center of the visual field (occupying roughly 30% to 80% of the frame).
- **Crowded Fields / Aggregates:** Images featuring dense clusters of multiple organisms or extensive bio-floc colonies will be extracted as a single contiguous foreground mask. This disrupts structural scale invariance and may cause misclassifications. In such cases, manually cropping the target individual prior to uploading is strongly advised.

4.4. Microscopy Modality

- **Illumination Mode:** The pre-trained weights were developed using standard optical brightfield RGB transmission microscopy.
- **Unsupported Modalities:** Distinct imaging modes such as fluorescence, darkfield, or differential interference contrast (DIC) feature inverted contrast and altered color spectra, which will undermine the statistical assumptions of the feature extractor.

Outcome be like:

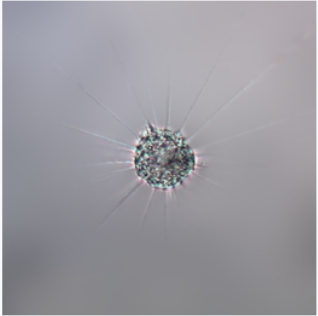
Two-Stage Intelligent EM Analysis System (EMDS-6)

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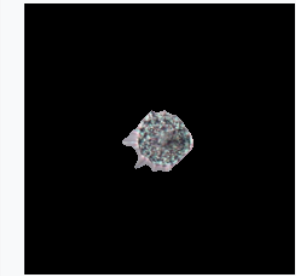
Environmental Microorganism Segmentation & Classification System

Open Microscopic Image

1. Raw Microscopic Slide



2. Stage 1 Segmented Foreground



3. Stage 2 Predictions & Confidence

Top 1: Actinophrys (99.81%)

Top 2: Stylongchia (0.03%)

Top 3: Gymnodinium (0.02%)

File: EMD55-g01-36.png
Stage 1: Background noise removed.
Stage 2: Features aligned for classification.

Completed | Predicted: Actinophrys

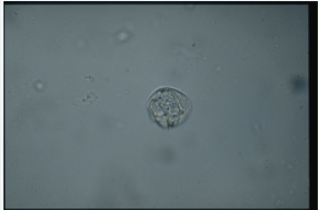
Two-Stage Intelligent EM Analysis System (EMDS-6)

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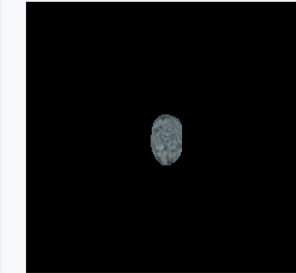
Environmental Microorganism Segmentation & Classification System

Open Microscopic Image

1. Raw Microscopic Slide



2. Stage 1 Segmented Foreground



3. Stage 2 Predictions & Confidence

Top 1: Aspidisca (99.30%)

Top 2: Synchaeta (0.15%)

Top 3: Gonyaulax (0.09%)

File: EMD55-g03-01.png
Stage 1: Background noise removed.
Stage 2: Features aligned for classification.

Completed | Predicted: Aspidisca